

Partial Results on the Generalized Two-Dimensional Sierpiński Problem

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Abstract

We study the first nontrivial two-prime case of Erdős–Graham Problem #203: whether there exists an integer $m \geq 1$, with $\gcd(m, 6) = 1$, such that

$$2^k 3^\ell m + 1$$

is composite for every pair $k, \ell \geq 0$. The central existence question remains open. This report records partial results toward the problem: covering-set criteria, density necessary conditions for both linear and nonlinear primes, an impossibility hierarchy for small covering moduli, the structural implication that every two-dimensional Sierpiński number is also a classical one-dimensional Sierpiński number, and computational evidence that the density condition is not sufficient for a two-dimensional covering. The latest WIT version incorporated here is WIT v10, which extends the main mixed-prime LCM obstruction to all primes $p \leq 50,000,000$.

1 Problem Statement and Status

The problem considered here is:

Does there exist $m \geq 1$, with $\gcd(m, 6) = 1$, such that

$$2^k 3^\ell m + 1$$

is composite for all $k, \ell \geq 0$?

This is the $r = 2$, $p_1 = 2$, $p_2 = 3$ case of Erdős–Graham Problem #203, which asks whether for given primes $p_1, \dots, p_r$ there is an integer m such that

$$p_1^{k_1} \cdots p_r^{k_r} m + 1$$

is composite for all $k_1, \dots, k_r \geq 0$.

Current status. The existence question remains open. No proof of existence and no proof of nonexistence is claimed in this report.

2 Definitions

Definition 2.1 (Linear prime). Let p be a prime with $\gcd(p, 6) = 1$. We call p *linear* if

$$3 \in \langle 2 \rangle \subseteq (\mathbb{Z}/p\mathbb{Z})^\times.$$

Equivalently, there exists $e \in \{0, \dots, \text{ord}_p(2) - 1\}$ such that

$$2^e \equiv 3 \pmod{p}.$$

In this case $e = \text{dlog}_2(3) \pmod{p}$ is defined relative to the cyclic subgroup $\langle 2 \rangle$.

Definition 2.2 (Diagonal strip). Let p be a linear prime, set $a = \text{ord}_p(2)$, and let $e = \text{dlog}_2(3) \pmod{p}$. For a residue $r \in \{0, \dots, a - 1\}$, define

$$S(p, r) = \{(k, \ell) \in \mathbb{Z}^2 : k + e\ell \equiv r \pmod{a}\}.$$

If m is coprime to p , then

$$p \mid 2^k 3^\ell m + 1$$

if and only if

$$(k, \ell) \in S(p, r_m), \quad r_m = \text{dlog}_2(-m^{-1} \pmod{p}).$$

Definition 2.3 (Subgroup attached to a prime). For every prime p with $\gcd(p, 6) = 1$, define

$$H_p = \langle 2, 3 \rangle \subseteq (\mathbb{Z}/p\mathbb{Z})^\times.$$

For fixed m coprime to p , put $c = -m^{-1} \pmod{p}$. Then

$$p \mid 2^k 3^\ell m + 1 \iff 2^k 3^\ell \equiv c \pmod{p}.$$

If $c \notin H_p$, the prime p covers no pairs (k, ℓ) for that m . If $c \in H_p$, the covered set has density $1/|H_p|$ on its period grid.

Definition 2.4 (Two-dimensional covering set). A finite set P of primes with $\gcd(p, 6) = 1$, together with compatible residue choices for each $p \in P$, is called a *two-dimensional covering set* for m if for every $k, \ell \geq 0$ some $p \in P$ satisfies

$$p \mid 2^k 3^\ell m + 1.$$

Such a covering set implies that m is a two-dimensional generalized Sierpiński number, provided each divisor is a proper divisor of $2^k 3^\ell m + 1$.

3 Basic Group-Theoretic Structure

Lemma 3.1 (Order of the generated subgroup). Let p be a prime with $\gcd(p, 6) = 1$. Let

$$a = \text{ord}_p(2), \quad b = \text{ord}_p(3).$$

Then

$$|H_p| = |\langle 2, 3 \rangle| = \text{lcm}(a, b).$$

Proof. The group $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of order $p - 1$. Let g be a primitive root modulo p , and write

$$2 \equiv g^\alpha, \quad 3 \equiv g^\beta \pmod{p}.$$

Then

$$a = \frac{p-1}{\gcd(\alpha, p-1)}, \quad b = \frac{p-1}{\gcd(\beta, p-1)}.$$

In the additive cyclic group $\mathbb{Z}/(p-1)\mathbb{Z}$, the subgroup generated by α and β is generated by $\gcd(\alpha, \beta)$. Hence

$$|\langle 2, 3 \rangle| = \frac{p-1}{\gcd(\alpha, \beta, p-1)}.$$

On the other hand, using the identity

$$\text{lcm}\left(\frac{n}{d_1}, \frac{n}{d_2}\right) = \frac{n}{\gcd(d_1, d_2)} \quad (d_1, d_2 \mid n),$$

with $n = p-1$, $d_1 = \gcd(\alpha, p-1)$, and $d_2 = \gcd(\beta, p-1)$, we get

$$\text{lcm}(a, b) = \frac{p-1}{\gcd(\alpha, \beta, p-1)}.$$

Thus $|H_p| = \text{lcm}(a, b)$. □

4 Density Lemmas

Lemma 4.1 (Diagonal strip density). *Let $a, L \geq 1$, with $a \mid L$, and let $e, r \in \{0, \dots, a-1\}$. Then the number of pairs $(k, \ell) \in \{0, \dots, L-1\}^2$ satisfying*

$$k + e\ell \equiv r \pmod{a}$$

is exactly L^2/a .

Proof. For each fixed ℓ , the congruence determines exactly one residue class for k modulo a , namely

$$k \equiv r - e\ell \pmod{a}.$$

Since $a \mid L$, the interval $\{0, \dots, L-1\}$ contains exactly L/a representatives of that residue class. Summing over all L choices of ℓ , the total number of solutions is $L(L/a) = L^2/a$. □

Lemma 4.2 (Subgroup density). *Let p be a prime with $\gcd(p, 6) = 1$, let*

$$H_p = \langle 2, 3 \rangle, \quad h = |H_p|,$$

and let $m \geq 1$ with $\gcd(m, 6) = 1$. Suppose

$$c \equiv -m^{-1} \pmod{p}$$

lies in H_p . If $L \geq 1$ and $h \mid L$, then the number of pairs $(k, \ell) \in \{0, \dots, L-1\}^2$ satisfying

$$p \mid 2^k 3^\ell m + 1$$

is exactly L^2/h .

Proof. Let $a = \text{ord}_p(2)$ and $b = \text{ord}_p(3)$. The map

$$\phi : \mathbb{Z}/a\mathbb{Z} \times \mathbb{Z}/b\mathbb{Z} \rightarrow H_p, \quad (k, \ell) \mapsto 2^k 3^\ell$$

is a well-defined surjective homomorphism. Since $|H_p| = h$, its kernel has size ab/h . Therefore each element $c \in H_p$ has exactly ab/h preimages modulo a and b .

Because $h = \text{lcm}(a, b)$, the condition $h \mid L$ implies $a \mid L$ and $b \mid L$. Each solution modulo a, b lifts to exactly

$$(L/a)(L/b)$$

pairs in the square $\{0, \dots, L-1\}^2$. Thus the number of pairs is

$$\frac{ab}{h} \cdot \frac{L}{a} \cdot \frac{L}{b} = \frac{L^2}{h}.$$

□

5 Covering Sets and Sierpiński Numbers

Theorem 5.1 (Covering implies compositeness). *Let P be a finite set of primes with $\gcd(p, 6) = 1$ for every $p \in P$, and let $m \geq 1$ satisfy $\gcd(m, 6) = 1$. Suppose that for all $k, \ell \geq 0$, there exists $p \in P$ such that*

$$p \mid 2^k 3^\ell m + 1.$$

If each such divisor is proper, then $2^k 3^\ell m + 1$ is composite for all $k, \ell \geq 0$.

Proof. Fix $k, \ell \geq 0$, and set

$$N = 2^k 3^\ell m + 1.$$

By hypothesis, some $p \in P$ divides N . If $1 < p < N$, then N has a proper prime divisor and is therefore composite.

In explicit CRT-derived covering constructions, the corresponding m is chosen so that $N > p$ for every prime in the covering set and every (k, ℓ) in the finite period grid. Equivalently, one verifies the finite small-case condition $N \neq p$ on the period grid. This is automatic in the large CRT constructions considered computationally below, where the resulting m is much larger than $\max P$. $\square$

Remark 5.2. The finite “proper divisor” check is not a density issue. It is a standard CRT verification step: after a proposed residue assignment is found, one checks on the finite period grid that the covering primes are proper factors of the covered values.

6 A Lovász Local Lemma Check

Lemma 6.1 (Failure of the symmetric Lovász local lemma test). *Let P_{5040} be the 31-prime set whose associated subgroup sizes $|H_p|$ all divide 5040, and let $N = 5040$. On the torus $\{0, \dots, N-1\}^2$, choose one residue class independently for each prime. For each cell (k, ℓ) , let $A_{k, \ell}$ be the event that no selected prime-residue class covers (k, ℓ) . The symmetric Lovász local lemma condition*

$$e \Pr(A_{k, \ell})(d + 1) \leq 1$$

fails by a large margin for this model.

Proof. For a fixed cell (k, ℓ) , prime p_i covers the cell with probability $1/h_i$, where $h_i = |H_{p_i}|$. Hence

$$\Pr(A_{k, \ell}) = \prod_i \left(1 - \frac{1}{h_i}\right).$$

The WIT computation records this probability as at least 0.36 for the 31-prime 5040-set.

The dependency degree is also enormous. Since prime p_i covers N^2/h_i cells, a fixed noncoverage event can interact with on the order of

$$\sum_i \frac{N^2}{h_i} = N^2 \sum_i \frac{1}{h_i}$$

other events. For the 5040-set,

$$\sum_i \frac{1}{h_i} = 1.021429\dots,$$

so the dependency degree is greater than 25,000,000. Therefore

$$e \Pr(A_{k,\ell})(d+1) > 2.7 \cdot 0.36 \cdot 25,000,000 \gg 1.$$

The symmetric local lemma therefore gives no existence guarantee for a full covering assignment. $\square$

Remark 6.2. This lemma is a negative result about a proof method only. It does not prove that the 5040-set has no covering assignment.

7 Density Necessary Conditions

Theorem 7.1 (Density condition for linear primes). *Let $P = \{p_1, \dots, p_n\}$ be a finite set of linear primes, and set*

$$a_i = \text{ord}_{p_i}(2).$$

If the corresponding diagonal strips cover all of $\mathbb{Z}_{\geq 0}^2$, then

$$\sum_{i=1}^n \frac{1}{a_i} \geq 1.$$

Proof. Let $L = \text{lcm}(a_1, \dots, a_n)$. Each strip for p_i occupies exactly L^2/a_i points in the square $\{0, \dots, L-1\}^2$, by the diagonal strip density lemma. Since the strips cover the whole square, the total number of strip incidences is at least L^2 . Therefore

$$\sum_{i=1}^n \frac{L^2}{a_i} \geq L^2.$$

Dividing by L^2 gives the result. $\square$

Theorem 7.2 (General density condition). *Let $P = \{p_1, \dots, p_n\}$ be a finite set of primes with $\gcd(p_i, 6) = 1$. Let*

$$h_i = |H_{p_i}| = |\langle 2, 3 \rangle \bmod p_i|.$$

Assume that for the fixed m under consideration the relevant target $-m^{-1} \pmod{p_i}$ lies in H_{p_i} for each i , and that the covering patterns

$$\{(k, \ell) : p_i \mid 2^k 3^\ell m + 1\}$$

cover all of $\mathbb{Z}_{\geq 0}^2$. Then

$$\sum_{i=1}^n \frac{1}{h_i} \geq 1.$$

Proof. Let $L = \text{lcm}(h_1, \dots, h_n)$. By the subgroup density lemma, the pattern corresponding to p_i covers exactly L^2/h_i points in $\{0, \dots, L-1\}^2$. Since the patterns cover the square,

$$\sum_{i=1}^n \frac{L^2}{h_i} \geq L^2.$$

Dividing by L^2 gives

$$\sum_{i=1}^n \frac{1}{h_i} \geq 1.$$

$\square$

Remark 7.3. For linear primes, $h_i = \text{ord}_{p_i}(2)$, so the general theorem recovers the linear density condition. For nonlinear primes, the subgroup size is

$$h_i = \text{lcm}(\text{ord}_{p_i}(2), \text{ord}_{p_i}(3)).$$

8 Impossibility Results

Theorem 8.1 (The four-prime set fails). *No two-dimensional covering set exists using only the linear primes*

$$\{5, 11, 23, 13\}.$$

Proof. The relevant orders are

$$\text{ord}_5(2) = 4, \quad \text{ord}_{11}(2) = 10, \quad \text{ord}_{23}(2) = 11, \quad \text{ord}_{13}(2) = 12.$$

Thus

$$\frac{1}{4} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12} = \frac{165 + 66 + 60 + 55}{660} = \frac{346}{660} < 1.$$

This violates the density necessary condition for linear primes. $\square$

Theorem 8.2 (Linear primes of order at most 60 fail). *No two-dimensional covering set exists using only linear primes p with $\text{ord}_p(2) \leq 60$.*

Proof. The linear primes with $\text{ord}_p(2) \leq 60$ are recorded as

$$5, 11, 23, 13, 19, 47, 29, 71, 37, 431, 97, 53, 59, 61,$$

with orders

$$4, 10, 11, 12, 18, 23, 28, 35, 36, 43, 48, 52, 58, 60.$$

Their reciprocal sum is

$$\begin{aligned} & \frac{1}{4} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12} + \frac{1}{18} + \frac{1}{23} + \frac{1}{28} + \frac{1}{35} \\ & + \frac{1}{36} + \frac{1}{43} + \frac{1}{48} + \frac{1}{52} + \frac{1}{58} + \frac{1}{60} = 0.81256769 \dots < 1. \end{aligned}$$

Every subset has still smaller reciprocal sum, so every such set violates the density necessary condition. $\square$

Theorem 8.3 (Linear-prime LCM obstruction). *Among linear primes $p < 5000$, no covering set whose order LCM is at most*

$$720720$$

can exist.

Proof. For every $L \leq 720720$, let

$$S_L = \{p < 5000 : p \text{ is linear and } \text{ord}_p(2) \mid L\}.$$

An exhaustive computation over $L \leq 720720$ gives

$$\max_{L \leq 720720} \sum_{p \in S_L} \frac{1}{\text{ord}_p(2)} = 0.7930527805527805 < 1.$$

Therefore every linear-prime set with LCM of orders at most 720720 violates the density necessary condition. $\square$

Theorem 8.4 (Mixed-prime LCM obstruction below 5040, computational scope). *For prime sets with all primes $p \leq 50,000,000$, no two-dimensional covering set can have*

$$\text{lcm}\{|H_p| : p \in P\} < 5040.$$

Proof. For $B < 5040$, define

$$\sigma_{\text{gen}}(B) = \sum_{\substack{p \leq 50,000,000 \\ \gcd(p,6)=1 \\ |H_p| \mid B}} \frac{1}{|H_p|}.$$

The computation recorded in the WIT file exhaustively sieves all primes $p \leq 50,000,000$. It finds that

$$\sigma_{\text{gen}}(B) < 1 \quad \text{for every } B < 5040.$$

The maximum below 5040 occurs at $B = 2520$, with value 0.917460. The Phase 6 extension found 10 new primes in the interval $(10^6, 50,000,000]$ with $|H_p| \mid 5040$, but their contribution is too small to make any $\sigma_{\text{gen}}(B)$ with $B < 5040$ reach 1.

If P had $\text{lcm}(|H_p| : p \in P) = B < 5040$, then

$$\sum_{p \in P} \frac{1}{|H_p|} \leq \sigma_{\text{gen}}(B) < 1,$$

contradicting the general density necessary condition. $\square$

Remark 8.5. This theorem has a computational scope: it is proved for primes $p \leq 50,000,000$. Primes $p > 50,000,000$ with $|H_p| \mid 5040$ would be large prime factors of $2^{5040} - 1$ and $3^{5040} - 1$; the WIT notes that a full universal closure would require factoring a very large integer or a new theoretical argument.

9 The Structural Reduction to the Classical Problem

Theorem 9.1 (Two-dimensional implies one-dimensional). *If $m \geq 1$, $\gcd(m, 6) = 1$, and*

$$2^k 3^\ell m + 1$$

is composite for all $k, \ell \geq 0$, then

$$2^k m + 1$$

is composite for all $k \geq 0$.

Proof. Set $\ell = 0$. Then

$$2^k 3^0 m + 1 = 2^k m + 1.$$

Thus the two-dimensional property immediately implies the classical one-dimensional Sierpiński property. $\square$

Corollary 9.2. *Every two-dimensional generalized Sierpiński number is a classical Sierpiński number.*

10 The Candidate $m = 78557$

The integer 78557 is the classical smallest known Sierpiński number. It does not solve the two-dimensional problem.

Theorem 10.1. *The integer $m = 78557$ is not a two-dimensional generalized Sierpiński number.*

Proof. At $(k, \ell) = (1, 2)$,

$$2^1 3^2 \cdot 78557 + 1 = 18 \cdot 78557 + 1 = 1414027.$$

The number 1414027 is prime, by the computational primality check recorded in the WIT evidence. Hence $2^k 3^\ell m + 1$ is not always composite for $m = 78557$. $\square$

Remark 10.2. Also $78557 = 17 \cdot 4621$, so $p = 17$ never divides

$$2^k 3^\ell \cdot 78557 + 1,$$

because the expression is always congruent to 1 (mod 17).

11 Computational Evidence

The following results are computational and should not be confused with a formal resolution of the open problem.

11.1 The 13-prime mixed set

Including nonlinear primes, the greedy density threshold is reached by the 13-prime set

$$\{5, 7, 11, 23, 13, 17, 19, 47, 29, 31, 71, 37, 73\}.$$

Its density is approximately

$$\sum_p \frac{1}{|H_p|} = 1.0056,$$

and

$$\text{lcm}(|H_p|) = 1275120.$$

However, the true two-dimensional verification domain is not $\text{lcm}(|H_p|)^2$. The recorded periods are

$$A = \text{lcm}(\text{ord}_p(2)) = 637560, \quad B = \text{lcm}(\text{ord}_p(3)) = 1275120,$$

so

$$AB = 812965507200 \approx 8.13 \cdot 10^{11}.$$

This makes exhaustive SAT/SMT verification currently infeasible. Thus the 13-prime set satisfies the density necessary condition, but its tileability remains unresolved.

11.2 The 31-prime 5040-set

A tractable set with $A = B = 5040$ is

$$P_{5040} = \{5, 7, 11, 13, 17, 19, 29, 31, 37, 41, 43, 61, 71, 73, 97, 113, 127, 181, \\ 211, 241, 281, 337, 421, 577, 631, 673, 1009, 2017, 2521, 3361, 13441\}.$$

For this set,

$$\sum_{p \in P_{5040}} \frac{1}{|H_p|} = 1.021429 > 1,$$

and the fundamental domain has

$$5040^2 = 25401600$$

cells.

Greedy set cover with 30 random restarts gave maximum coverage

$$18216600/25401600 = 71.714\%.$$

The number of uncovered cells was

$$7185000,$$

which is exactly

$$\frac{7185000}{25401600} = \frac{99}{350}.$$

No full covering assignment was found.

Remark 11.1. This is strong evidence that the density condition is not sufficient for two-dimensional covering. Strictly speaking, greedy failure alone is not a formal proof that no assignment exists, but later ILP and simulated annealing attempts also failed to find a covering.

11.3 ILP, simulated annealing, and Lovász local lemma checks

For the 5040-set, an ILP formulation used binary variables

$$y_{p,r} \in \{0, 1\},$$

where $y_{p,r} = 1$ means that prime p uses residue r . The recorded model had 9279 binary variables and thousands of seed covering constraints. CBC did not find an integer feasible solution in the recorded runs, but did not produce a formal infeasibility certificate.

Simulated annealing with $1.5 \cdot 10^6$ iterations reached

$$71.570\%$$

coverage, below the greedy maximum 71.714%.

The formal WIT v10 lemma *lovasz-lll-fails* records the corresponding cell-event local lemma calculation. For a fixed cell (k, ℓ) , the noncoverage event has probability

$$\prod_i \left(1 - \frac{1}{h_i}\right),$$

and the dependency degree is on the order of

$$5040^2 \sum_i \frac{1}{h_i} > 25,000,000.$$

Consequently the symmetric LLL condition fails by many orders of magnitude.

11.4 The 10080-modulus search

The next tractable modulus considered was

$$N = 10080 = 2 \cdot 5040.$$

The corresponding search used 33 primes, with density

$$1.031944.$$

The best greedy coverage was

$$72.07\% = 73223308/101606400.$$

No covering was found. The marginal improvement from 5040 to 10080 was only about 0.36 percentage points.

11.5 Phase 6: extension to $p \leq 50,000,000$

The Phase 6 computation extended the T6b sieve from $p \leq 10^6$ to

$$p \leq 50,000,000.$$

It scanned 3,932,436 primes in $(10^6, 50,000,000]$, testing

$$2^{5040} \equiv 1 \pmod{p}, \quad 3^{5040} \equiv 1 \pmod{p}.$$

It found 10 new primes with $|H_p| \mid 5040$:

$$\begin{aligned} &2035153, 2543689, 3391249, 5084857, 6779137, \\ &8473081, 13561969, 27110497, 33884761, 45175201. \end{aligned}$$

Their subgroup sizes are

$$1008, 2520, 5040, 5040, 5040, 5040, 5040, 5040, 5040, 1680,$$

respectively. The total density increment is

$$0.003373.$$

Thus

$$\sigma_{\text{gen}}(5040) = 1.024802$$

for the 41 primes found up to 50,000,000, while

$$\sigma_{\text{gen}}(B) < 1 \quad (1 \leq B < 5040).$$

The maximum below 5040 remains at $B = 2520$, now with value

$$\sigma_{\text{gen}}(2520) = 0.917460.$$

The same Phase 6 run also performed a greedy and local-search covering attempt on a 23-prime subset of density 0.944. The best coverage was

$$17683759/25401600 = 69.62\%,$$

and no covering was found.

12 Summary of Main Contributions

1. The problem is identified as the $r = 2$, $(p_1, p_2) = (2, 3)$ case of Erdős–Graham Problem #203.
2. A general density necessary condition is proved: every finite two-dimensional covering set must satisfy

$$\sum_{p \in P} \frac{1}{|H_p|} \geq 1, \quad H_p = \langle 2, 3 \rangle \bmod p.$$

3. The subgroup size is proved generally:

$$|H_p| = \text{lcm}(\text{ord}_p(2), \text{ord}_p(3)).$$

4. Several families of covering sets are ruled out by density: the four-prime set $\{5, 11, 23, 13\}$, all linear primes with $\text{ord}_p(2) \leq 60$, linear-prime sets with LCM at most 720720, and mixed-prime sets with $p \leq 50,000,000$ and $\text{lcm}(|H_p|) < 5040$.
5. Any two-dimensional generalized Sierpiński number is necessarily a classical one-dimensional Sierpiński number.
6. The natural candidate $m = 78557$ is ruled out by the prime value

$$2^1 3^2 \cdot 78557 + 1 = 1414027.$$

7. Computational evidence shows that density greater than 1 is not by itself enough to produce a two-dimensional covering: the 31-prime 5040-set has density 1.021429, but extensive search reached only about 71.7% coverage.
8. Phase 6 extends the $B^* = 5040$ computation to all primes $p \leq 50,000,000$: $\sigma_{\text{gen}}(B) < 1$ for all $B < 5040$, while $\sigma_{\text{gen}}(5040) = 1.024802$.

13 Version History of the WIT Development

Version	Commit	Steps/Lines	Main addition
v1	19d70c6	20 / 242	Linear-prime framework
v2	c04460f	31 / 455	General H_p theorem
v3	db4a6b3	34 / 596	5040-set SAT evidence
v4	05d0ed3	34 / 648	Uncovered-structure analysis
v5	22598e6	38 / 702	Cyclic-subgroup proof and T1 fix
v6	3fbf6ba	42 / 940	T6b and GAP4 evidence
v6-final	de7ca08	42 / 1005	$N = 10080$ evidence
v7	f482a25	42 / 1069	Scope and Lovász documentation
v8	caacb14	42 / ~ 1120	T6b extended to $p \leq 10^6$
v9	b11f534	45 / ~ 1260	Formal Lovász lemma and verification note
v10	ec217a3	45 / ~ 1340	T6b extended to $p \leq 50,000,000$

14 WIT v10 Status

The current WIT artifact is recorded as WIT v10, commit `ec217a3`. It contains 45 top-level WIT steps and approximately 1340 lines. Its structural contents are:

Block type	Contents
Definitions	linear-prime, diagonal-strip, 2d-covering-set, subgroup-prime
Lemmas	cyclic-subgroup-order, diagonal-strip-density, subgroup-density, lovasz-lll-fails
Theorems	covering, density, impossibility, T6b, 2D-to-1D, $m = 78557$
Conjecture	two-dimensional Sierpiński existence, marked STATUS OPEN
Evidence	5040, 10080, T6b universal, and Phase 6 computations

The WIT checker output recorded in the file is:

45 top-level steps, 0 sub-steps, all references valid.

The same file also includes a human-verifiable checklist for readers who do not have the WIT checker available. A fresh critic session for WIT v10 reported “complete, ship” after checking the five requested items: open status of the main conjecture, T6b scope $p \leq 50,000,000$, the Phase 6 evidence block, the formal Lovász lemma, and the citation of the small-case remark in the covering theorem.

15 Open Problems

Conjecture 15.1 (Two-dimensional generalized Sierpiński existence). *There exists $m \geq 1$, with $\gcd(m, 6) = 1$, such that*

$$2^k 3^\ell m + 1$$

is composite for all $k, \ell \geq 0$.

This conjecture remains open. The present work provides obstructions and computational evidence, but neither constructs such an m nor proves that no such m can exist.

References

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